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Circuit and working

The circuit described here (refer Fig. 3) lets you control an electromagnetic relay so that it turns on a white LED light when the preset light level is reached and turns it off when a different preset level is reached. The circuit is built around NE555 IC (IC1). The 5mm light dependent resistor (LDR1) in the circuit triggers the 12V electromagnetic relay (RL1) as per the ambient light level. RL1 drives the 10mm white LED light source (LED2). Series resistor (R2) is included to limit the white LED current.

Note that switching threshold is determined by a 470k potentiometer (VR1) that causes the output to toggle with the preset threshold values. The light source (LED2) automatically switches on when it gets dark and switches off when there is sufficient ambient light. The 100µF capacitor (C1) provides a bit of hysteresis to prevent the circuit from jittering near the threshold level. The circuit is optimised for use with a nominal DC voltage of 12V drawn from any standard accumulator commonly used in boats.

Construction and testing

An actual-size, single-side PCB pattern for the anchor light circuit is shown in Fig. 4 and its component layout in Fig. 5.

The circuit assembled on the small PCB can fit easily inside most prototype/custom enclosures, which should be waterproof for mounting on the masthead. If possible, try to add some optics (lens and reflector) with the white LED (LED2) to spread the light outward. The 12V power supply input wires can then be connected to corresponding wires extending from the existing electric-points of the anchor light. Fig. 6 shows how the prototype may be assembled. Author's prototype is shown in Fig. 7.

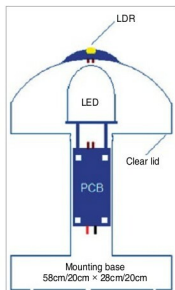


Fig. 6: Suggested enclosure layout



Fig. 7: Author's prototype



T.K. Hareendran is founder and promoter of TechNodo Protolabz